Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	SKILLWALLET-EV-01
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python 3.11, SQL, HTML5, CSS3	Data validation, Flask web framework routing, and frontend dashboard embedding.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV (ev_market_test.csv), Tableau Data Extract (.hyper)	Stores 100,000 EV data points; Hyper extract enables ultra-low latency query execution.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Text File Connector & Python Pandas	Direct structured ingestion and schema validation across 37 vehicle attributes.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop Calculated Fields & Python Scripts	Engineered KPIs: Cost per Km Range, Degradation Risk index, and Total Annual CO2 Emissions.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop & Tableau Public	Developed 9 interactive analytical visualisations, consolidated dashboards, and storyboard.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and	Flask Web App, Jinja2, Tableau Public Embedded API	Delivers responsive web interface with seamless drill-down and

		visual insights.		cross-filtering for end users.